PHASE 6 DOCUMENT

ComicCraft: AI Comic Story Creator using Gemini Models

Phase Name: User Management, Content Safety and Collaboration Features

Project Name	ComicCraft: AI Comic Story Creator using Gemini Models
Phase	Phase 6 – User Management, Safety and Collaboration
Project Type	Generative AI Comic Creation Application
Main Technology	Google Gemini Models
Prepared By	_____
Guide / Faculty	_____
Date	_____

Academic Project Phase Documentation

1. Phase Overview

This is Phase 6 of **ComicCraft: AI Comic Story Creator using Gemini Models**. This phase strengthens the platform around the creative tools developed earlier. It focuses on user accounts, privacy controls, content-policy enforcement, reporting and moderation processes, administrative management, and optional sharing or collaboration features for creators who choose to work together.

2. Phase Objective

The objective of this phase is to make ComicCraft a safer and more manageable platform for AI-assisted storytelling. Users should be able to control who can view their comic projects, administrators should be able to manage access and policy settings, and the system should provide clear procedures for handling prohibited or inappropriate content. Any collaboration capability should be permission-based and should protect the original creator's work.

3. User Account Management

User Account Management handles registration, login, logout, password protection, profile updates, account recovery, and account deletion requests. Each account should have a unique identifier and a defined role. Creators can manage their own profile preferences, project settings, and privacy choices. The system should store only the data necessary to provide the service and should explain how user data and comic drafts are handled.

4. Roles and Permissions

ComicCraft supports at least two roles: Creator and Administrator. Creators can create, edit, save, export, share, or delete their own comic projects. Administrators can manage users, usage limits, templates, content categories, reports, and platform policies. Future roles may include Educator, Reviewer, or Collaborator. Role-based access control ensures that a user sees only the actions and information permitted for that role.

5. Project Privacy Controls

Each comic project should have a privacy setting such as Private, Shared with Specific Users, or Public Preview where the platform supports publishing. Private is the default option. A creator can choose to share a project only with named collaborators or selected reviewers. The system must verify permissions before displaying project content, edits, exports, or version history. Changing a privacy setting should require a clear user action and should be recorded in the project activity history.

6. Optional Collaboration Features

Collaboration can allow a creator to invite another registered user to contribute to a comic project. A project owner may assign permissions such as Viewer, Commenter, or Editor. Viewers can read shared material; commenters can provide feedback; editors can modify approved portions according to the owner's rules. Collaboration should clearly show who made each change, retain version history, and allow the owner to revoke access at any time. This feature can be implemented incrementally after core private-project security is verified.

7. Content Safety Policy

ComicCraft should define clear content rules. The system should prohibit requests or output involving illegal activity, sexual content involving minors, hateful or discriminatory material, harmful instructions, explicit violence beyond allowed settings, impersonation intended to deceive, or infringement-focused requests. The policy should also discourage copying copyrighted characters, logos, or living artists' distinctive styles in a misleading manner. The application should state that AI-assisted content must be reviewed by the user before sharing or publishing.

8. Content Controls and Safe Responses

Input controls and Gemini safety settings should be used to detect or refuse prohibited requests. When a request cannot be completed, ComicCraft should respond politely and, where possible, offer a safe alternative such as creating an original character or family-friendly scene. The application should avoid exposing internal moderation details or presenting unsafe content to users. Safety handling should be consistent across initial generation, targeted regeneration, dialogues, captions, exports, and shared projects.

9. Reporting and Moderation Workflow

Users should be able to report content, a shared project, or platform behavior that may violate policy. A report can include the project reference, reason category, optional explanation, reporter identity where required, and date. Administrators review reports according to defined guidelines and may take actions such as no action, warning, content restriction, removal from public preview, temporary account limitation, or account suspension. Moderation actions should be documented and access should be limited to authorized administrators.

10. Admin Management Features

The administrator interface can manage account status, roles, content categories, prompt templates, feature flags, usage quotas, report queues, and policy notices. Administrators should not casually browse private creator content; access should be restricted to legitimate operational or moderation needs and recorded in audit logs. Administrative actions that affect accounts, policies, or content visibility should be traceable and reversible where possible.

11. Privacy and Data Protection

ComicCraft should collect the minimum personal information needed for account operation. Project content, prompts, generated drafts, collaboration data, and logs should be protected using secure access controls. Users should be informed about retention and deletion options. The application should avoid using private comic content for unrelated purposes. Sensitive credentials, including Gemini API keys, must remain server-side and inaccessible to creators or collaborators.

12. Audit and Activity History

An activity history can record significant project events: creation, edit, regeneration, share invitation, permission change, export, privacy change, and deletion. In collaboration, each edit should show the contributor and time. Audit records support accountability, help recover from mistakes, and assist with moderation or technical investigations. Access to detailed audit information should be restricted according to user role and privacy settings.

13. Security Controls

Security controls include secure authentication, hashed passwords, session expiry, server-side authorization checks, input validation, output encoding, HTTPS communication, least-privilege access, secure secret storage, rate limits, and audit logs. The system should prevent a creator from accessing another user's private projects by modifying a project identifier. It should also protect collaboration invitations from unauthorized acceptance or reuse.

14. Testing Plan

Testing should verify registration and login, role-based menus, private-project isolation, sharing invitations, collaborator permissions, access revocation, activity-history accuracy, report submission, administrator report review, content-policy refusals, safe alternative responses, account-status restrictions, and data-deletion behavior. Security tests should confirm that unauthorized users cannot retrieve projects, exports, or drafts through direct URL or API requests.

15. Phase Deliverables

The deliverables of Phase 6 are user-account management, role and permission rules, project privacy settings, collaboration design or initial features, content-safety policy, safe-response handling, reporting and moderation workflow, admin management interface, privacy controls, audit-history design, security controls, and test evidence for access and safety features.

16. Planned Next Phase

The next phase will be **Phase 7: System Testing, Quality Assurance and Security Validation**. It will test the complete ComicCraft application end-to-end, document test cases and outcomes, validate Gemini integration and content safety, confirm privacy and access controls, identify defects, and prepare the system for final deployment.

17. Conclusion

Phase 6 makes ComicCraft more than a content-generation tool by adding the controls required for responsible creative use. Account management, privacy settings, content safety, moderation, and optional collaboration protect creators and the platform. With these controls in place, ComicCraft is ready for full quality assurance and security validation.

Phase Approval

Prepared by	_____
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Approved by	_____